

Research

A novel CNN-GRU-LSTM based deep learning model for accurate traffic prediction

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Received: 15 October 2024 / Accepted: 27 March 2025

Published online: 15 April 2025

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Abstract

Effective traffic prediction is crucial for optimizing urban transportation systems, minimizing congestion, and enhancing overall efficiency. Traffic congestion results in prolonged travel durations, higher fuel consumption, economic setbacks, and increased environmental pollution. To tackle these issues, we introduce a Hybrid CNN-GRU-LSTM model—an advanced deep learning framework that combines convolutional neural networks (CNN), gated recurrent units (GRU), and long short-term memory (LSTM) networks. This integrated model is specifically designed to capture both spatial and temporal patterns of traffic flow, making it highly effective for predicting vehicle volumes at intersections. The Hybrid CNN-GRU-LSTM leverages CNN to model spatial dependencies between road segments, while GRU and LSTM layers handle short-term and long-term temporal patterns in traffic data. This combination allows for more accurate predictions by incorporating spatial relationships and temporal dynamics simultaneously. The model was tested using publicly available datasets, including PeMS, the England dataset, the P/Castellano dataset, and the Fedesoriano dataset, and results demonstrate that Hybrid CNN-GRU-LSTM significantly outperforms several state-of-the-art models, achieving a reduction of up to 30–35% in error values. This study highlights the effectiveness of combining CNN, GRU, and LSTM architectures for traffic prediction, offering a robust solution for transportation management. The proposed model's significant improvement in prediction accuracy can help mitigate the adverse effects of traffic congestion and enhance the overall performance of transportation networks.

Keywords Traffic congestion · Traffic prediction · Long short-term memory (LSTM) · Gated recurrent unit (GRU) · Convolutional neural network (CNN)

1 Introduction

Increasing impact of traffic on transportation and traffic management systems has become a critical concern for urban planning. As cities grow and populations rise, the demand for efficient transportation systems and effective traffic management strategies has intensified. Congestion is a significant consequence of increasing traffic, leading to longer travel times, reduced fuel efficiency, higher pollution, and greater risks of accidents and fatalities.

Traditional traffic prediction methodologies, such as statistical models, often struggle to effectively capture the intricate and dynamic nature of real-time traffic patterns. In contrast, deep learning architectures, particularly hybrid models, offer substantial enhancements in this domain. The Hybrid CNN-GRU-LSTM model integrates convolutional neural networks (CNNs), gated recurrent units (GRUs), and long short-term memory (LSTM) networks to adeptly manage the

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spatial–temporal complexities inherent in traffic data. This integration enables the model to proficiently extract spatial features through CNNs, model temporal dependencies via GRUs, and retain long-term temporal information using LSTMs, thereby improving the accuracy and robustness of traffic flow predictions. For instance, a study demonstrated that a hybrid CNN-GRU model outperformed traditional methods in traffic flow forecasting by effectively capturing both spatial and temporal dependencies.

CNNs capture spatial relationships between road segments, identifying how vehicle flow is interconnected across different areas. GRU and LSTM layers model the temporal aspects, learning from historical data to predict future traffic conditions at road junction based on short-term fluctuations and long-term trends. This hybrid model ensures a more accurate prediction of traffic flow than traditional models, which typically struggle with the dynamic and multi-layered nature of traffic.

By leveraging the power of deep learning, Hybrid CNN-GRU-LSTM enables more reliable real-time traffic predictions, allowing for better congestion management, enhanced road safety, and more efficient urban transportation systems. This predictive accuracy not only improves daily traffic flow but also aids long-term infrastructure planning, helping cities meet the growing demands of urbanization.

Efficient traffic management depends on minimizing traveller waiting times at junctions and optimizing their flow through the road network. Traffic prediction at junction forecasts vehicle movement at specific intersections to estimate the number of vehicles, their movements, and traffic timing. Advanced technologies and data analysis techniques, such as historical data analysis, sensor and camera networks, and machine learning models, are employed to achieve accurate predictions. Deep learning models such as the dynamic graph convolution neural network [1] and geometric feature extraction based on the dynamic convolutional neural network [2] have been adopted to address varied engineering problems. In this context, deep learning is also gaining attention for smart city solutions, and a recent survey [3] explores its applications, progress, challenges, and opportunities to inspire experts and new researchers in the field.

These solutions promote and enhance public transportation networks, encouraging the use of mass transit and reducing the reliance on private vehicles. Additionally, promoting the expansion of road infrastructure and alternative transportation modes, such as cycling and walking, can help alleviate traffic congestion. However, these solutions can only be implemented with a major breakthrough in the regulatory mechanisms and a change in the mindset of people at large. Until such a change happens, it is evident that an effective traffic management system must be developed and implemented to tackle the issues caused by rising traffic congestion.

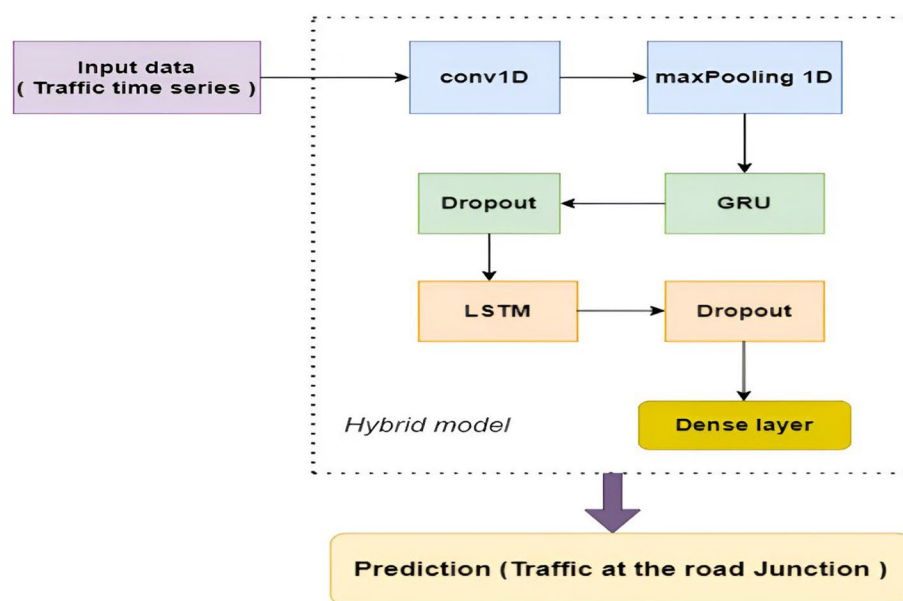
The modern city is transforming into an intelligent city as urbanization and population expansion imposes significant challenges on urban traffic control. An Intelligent Transportation System (ITS) employs advanced technologies to enhance traffic management. It accomplishes this by utilizing real-time monitoring, adjusting traffic signals dynamically, and providing adaptive route guidance. An ITS is a crucial element of a well-planned smart city, with the prediction of traffic on a particular road as a crucial ITS constituent. Precise traffic prediction plays a vital role in a wide range of real-life scenarios, including reducing congestion and enabling vehicle demand prediction. The continuously increasing availability of traffic-related data offers exciting opportunities to investigate this problem from additional perspectives.

Researchers use intelligent transportation approaches to estimate traffic flow using statistical models, which involves substantial work owing to established assumptions that may result in inaccurate predictions. Deep neural models such as recurrent neural networks (RNNs) may enhance prediction accuracy by remembering past data. However, because of the vanishing gradient problem, RNNs may lose the memory of more extensive sequences over time [2]. Researchers have suggested numerous RNN modifications to overcome this problem. LSTM and GRU are commonly employed. The GRU model, first introduced by [3], tries to avoid the frequent vanishing gradient issue in traditional recurrent neural networks.

In this research, we focus on predicting traffic flow. A deep learning model is presented that combines a hybrid architecture of CNN, GRU, and LSTM to forecast the number of vehicles passing through a particular junction, as illustrated in Fig. 1. The study sheds light on the hybridization of these three techniques, presenting the Hybrid CNN-GRU-LSTM model and showcasing its efficacy in traffic flow prediction tasks. Furthermore, the research provides an overview of traffic prediction at junctions and explores the implications of increasing traffic on transportation and traffic management systems. Integrating CNN, GRU, and LSTM architectures into a hybrid deep learning model can effectively capture the spatial and temporal dynamics of traffic flow, leading to significantly improved traffic prediction accuracy compared to state-of-the-art methods.

The research begins with exploring the dataset, followed by data pre-processing and feature engineering to prepare the data for training, subsequent to which the models are applied. The main contributions of our research work are as follows:

Fig. 1 Traffic prediction at junction using Hybrid CNN-GRU-LSTM



- To propose a hybrid deep learning architecture, namely a Hybrid CNN-GRU-LSTM model for traffic flow prediction.
- To apply the model on four publicly available datasets and present the results.
- To establish, by the results of experimentation, the effectiveness of multi-layer models for spatial feature mining and for handling sequence data.

In Sect. 2, the related work reviews existing research and approaches in traffic flow prediction, highlighting the need for a hybrid model for efficient traffic management. Section 3 presents the proposed hybrid model architecture, including the combination of machine learning techniques, and explains the rationale behind selecting specific hyperparameters. Section 4 describes and provides an exploratory analysis of the datasets used in this study. Moving on to Sect. 5, the experimental setup and performance parameters are explained. Section 6 evaluates the performance of the hybrid model using error metrics like Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). The part also contains a comparison study, illustrating the hybrid model's superiority over other machine learning models in properly forecasting traffic flow. Finally, Sect. 6 summarizes the results, stressing the contributions to traffic prediction and recommending future research avenues for improving the hybrid model and incorporating it into real-time traffic systems.

2 Related work

Traditional traffic prediction methods, such as statistical models (e.g., ARIMA) and time-series forecasting, fall short of capturing the complex, non-linear, and high-dimensional nature of real-world traffic data. These methods struggle to model both spatial and temporal dependencies, limiting their accuracy and generalization across different traffic conditions. In contrast, deep learning and hybrid models, like Hybrid CNN-GRU-LSTM, offer superior performance by effectively handling the spatial-temporal interactions within traffic data. Convolutional Neural Networks (CNNs) extract spatial features, while GRU and LSTM layers capture short-term and long-term temporal dependencies, respectively. This integration improves the model's capacity to capture complex patterns, adapt to dynamic conditions, and support real-time applications in intelligent transportation systems. With access to large datasets and the ability to continuously learn from streaming data, deep learning models provide more accurate, flexible, and scalable traffic predictions, outperforming traditional methods in modern traffic control and management systems. Below, we discuss the related work based on these approaches and explain why we chose hybridization to improve prediction performance.

2.1 Traditional machine learning and statistical approaches for traffic prediction

Traffic prediction is critical in traffic management systems because it may assist in optimizing resource utilization, minimizing congestion, and improving safety. Recent studies on traffic prediction utilizing deep learning techniques have been undertaken. Several recent surveys have examined traffic prediction literature from various perspectives in specific contexts. A survey conducted between 2004 and 2013 reviewed various methods and applications and identified significant challenges during that time period. It primarily focuses on short-term traffic prediction and relies mainly on traditional methods [4]. Another research [5] also prioritized short-term traffic prediction and provided an overview of prediction techniques and research recommendations. In addition, the paper [6] supplied information on obtaining road traffic data. To estimate traffic congestion, a predictive model was presented that used real-time data from a traffic monitoring system and multiple machine learning techniques [7], such as Gradient Boosting (GB), Support Vector Machine (SVM), and Random Forest (RF). The authors evaluate the performance of the suggested system by comparing the prediction accuracy of the different machine-learning algorithms.

Predicting traffic congestion levels using Extreme Learning Machines (ELMs), which can learn from new data on the fly without retraining the entire model, has been investigated [8]. This method is compared with other traditional approaches, demonstrating that the proposed model outperforms them in terms of prediction accuracy. The authors also experimented with different hyperparameters to optimize the model's performance.

Traditional approaches to traffic prediction comprise statistical techniques. The statistical process involves developing a model from data for making predictions, utilizing machine learning models ARIMA [9], and VAR [10]. However, these methods have specific requirements for traffic data, which can take time to meet when dealing with complex and dynamic traffic data. Additionally, these traditional methods are used to evaluate and analysis small datasets. Several conventional machine learning methods like Support Vector Regression [11] and Random Forest Regression [12] have been suggested for traffic prediction to address these limitations.

2.2 Deep learning application in traffic flow prediction

LSTM and GRU are advanced deep learning models designed to handle sequence-based prediction tasks, such as traffic flow forecasting [13]. Both LSTM and GRU are specialized recurrent neural networks (RNNs) that capture long-term dependencies in time-series data, making them highly effective for predicting future traffic patterns based on historical data. LSTMs use a memory cell controlled by three gates—input, output, and forget gates—that regulate the flow of information, enabling the model to retain or discard information as needed. This structure allows LSTMs to overcome the vanishing gradient problem in traditional RNNs, making them particularly useful for capturing long-term patterns in traffic data, such as recurring congestion during rush hours or seasonal fluctuations.

GRUs, on the other hand, are a simpler alternative to LSTMs. They combine the input and forget gates into a unified 'update gate', reducing the complexity of the model while maintaining similar performance. GRUs are computationally more efficient than LSTMs and often perform comparably, especially in tasks where shorter-term dependencies are more prominent. In traffic prediction, both LSTM and GRU excel at learning the temporal dynamics of vehicle flow, allowing them to make highly accurate forecasts. These models have demonstrated significant improvements in traffic prediction accuracy compared to traditional time-series methods, leading to better real-time traffic management, reduced congestion, and improved long-term urban planning. Their ability to process both immediate variations and extended trends makes them ideal for solving complex transportation problems.

Meanwhile, [14] provided a comprehensive survey of traffic prediction using machine learning techniques, including at intersections. The authors discussed the different approaches used to process traffic data and highlighted the strengths and weaknesses of each approach. The traffic flow prediction model introduced in [15] utilizes graph convolutional neural networks to enhance intersection traffic control. By employing actual traffic data from real-world scenarios, they substantiated the efficiency of their model. This research expands in [16] to propose a traffic flow prediction model that uses deep learning with an attention mechanism to improve intersection control.

A short-term algorithm for traffic flow prediction improves junction management by combining deep learning with spatial and temporal data. Their approach considers geographical and temporal correlations in traffic data, resulting in more accurate forecasts. Meanwhile, [17] suggested a short-term traffic flow prediction model that improves intersection management using a convolutional neural network using spatial and temporal information. They tested the model's efficiency by using traffic data from real-world circumstances.

A traffic flow prediction algorithm incorporating deep learning and principal component analysis to enhance junction control [18] considers the nonlinear connection between traffic data and successfully forecasts traffic flow. The author of [19] suggested a model combining CNN (Convolutional Neural Network) and short-term traffic flow prediction with a spatial–temporal attention mechanism. Their approach includes traffic data and geographical and temporal connections, resulting in more accurate forecasts. A traffic flow prediction model that improves intersection management by combining a convolutional neural network with transfer learning [20]. Their approach improves prediction accuracy by using information gathered from pre-trained models.

In [21], the authors provide an overview of using deep learning to offer real-time data to commuters and transportation authorities, aiding them in reducing traffic through effective forecasting. The authors emphasize the importance of accurate traffic prediction for improving transportation management, reducing congestion, and improving safety when using deep learning techniques for traffic prediction, such as CNN, RNN, and their variants, such as LSTM and GRU.

A deep learning model [22] is used to predict traffic flow using an LSTM encoder-decoder architecture, considering historical traffic flow data and weather conditions as inputs to predict future traffic flow. The model comprises an LSTM encoder to capture temporal dependencies in the input sequence and a decoder to generate predictions of future traffic flow. The model is trained using a mean squared error loss function and optimized with the Adam optimizer.

Most of the deep learning models are used for predicting traffic flow prediction and assessment; LSTM is one of the techniques in which encoder-decoder architecture is proposed [23]. The authors train the model on historical traffic flow data and use it to predict the traffic flow for the next 30 min to several hours. They compare their method with other traditional methods, and the present LSTM model provides better performance. The authors also experiment with different parameters to optimize the models performance.

Authors in [24] develop a hierarchical model that can directly map the input to output. This is achieved using deep learning models comprising several learnable blocks or layers. These layers are stacked on top of each other to form a deep architecture. Additionally, CNN [25], RNN [26, 27] and their variations, such as LSTM [28] and GRU [29], have also been employed in deep learning models and are particularly useful in handling sequential data where there is a dependence on previous inputs. For instance, in the context of traffic prediction for intelligent transportation systems [30], such models can effectively capture spatiotemporal dependencies, enabling accurate forecasting of traffic flow, congestion levels, and travel times.

2.3 Machine learning approaches and advancement in traffic prediction for ITS

Traffic prediction using machine learning, as discussed in 'Traffic Prediction for Intelligent Transportation Systems (ITS) Using ML [33], examines the application of machine learning in predicting traffic patterns within the realm of intelligent transportation systems (ITS). The authors employ machine learning algorithms to analyze historical traffic data and predict future patterns. Their results demonstrate the efficacy of machine learning in enhancing traffic prediction accuracy, which is crucial for optimizing traffic management in ITS.

The paper [34] focuses on traffic flow prediction, leveraging machine learning techniques to investigate the ability of algorithms to forecast traffic flow dynamics. The research work provides important insights into the accuracy of different predictive models by analyzing the temporal and spatial dependencies in traffic patterns. Meanwhile, the rapid growth of big data and IoT has outpaced processor performance, creating significant energy consumption and efficiency challenges in large-scale systems like cloud data centres. To address these issues, another paper [35] explores energy and performance challenges, surveys state-of-the-art methods, and identifies key research areas for sustainable computing.

2.4 Machine learning application in connected and autonomous vehicles and vehicle Ad Hoc network

In [36], the focus shifts towards integrating machine learning into the connected and autonomous electric vehicles (CAEVs) context. The authors propose a model that predicts traffic conditions to optimize the speed of CAEVs, contributing to energy efficiency and overall performance.

The fourth paper [37] extends the scope to Vehicular Ad Hoc Networks (VANETs), where the authors propose a network traffic prediction model. By considering road traffic parameters, the study employs artificial intelligence methods to enhance the accuracy of traffic predictions within VANETs. The integration of AI methods underscores the adaptability of machine learning techniques to diverse transportation scenarios.

2.5 Traffic prediction using graph neural network

In the context of smart cities, predicting traffic flow remains a important challenge due to the complex interplay of spatio-temporal factors and non-linear dynamics. Traditional approaches often struggle with high computational costs and fail to effectively identify patterns, particularly overlooking relationships between far-apart roads. In contrast, the graph convolutional network (GCN) [31] proves to be better suited for handling graph-structured data, like the architecture of traffic networks.

The authors in [32] tackle the challenges associated with traffic flow networks by proposing innovative methods. One notable approach involves the use of 3D temporal graph convolutional networks (3D-TGCN) for traffic prediction. By constructing road graphs based on time series similarity, the need for spatial information becomes redundant. The model's unique 3D graph convolution technique significantly improves accuracy, outperforming existing baseline models in empirical evaluations. Likewise, traffic forecasting faces difficulties in capturing intricate spatio-temporal relationships and non-linear patterns. Many current methods fail to directly integrate spatio-temporal dependencies, instead handling them separately.

The adaptive graph spatial-temporal transformer network (ASTTN) represents a significant advancement inspired by the success of transformer architectures. By leveraging local multi-head self-attention mechanisms on an adaptive spatial-temporal graph, ASTTN effectively captures cross-spatial-temporal correlations and delivers outstanding empirical results [33]. Real-world traffic prediction remains challenging due to the dynamic nature of spatiotemporal dependencies in traffic data. To address this, the adaptive spatio-temporal convolutional network (ASTCN) is proposed, integrating a spatial graph learning module and an adaptive temporal convolution module. Comprehensive experiments on real-world traffic datasets demonstrate that ASTCN consistently surpasses state-of-the-art approaches [34].

The complexity of spatial-temporal correlations within road networks further complicates traffic forecasting. The Dual-Graph framework mitigates this issue by simultaneously modelling traffic dynamics for both nodes and edges, with evaluations on public datasets highlighting its effectiveness, particularly in long-term forecasting scenarios [35]. In fields like transportation and neurology, spatiotemporal forecasting becomes even more complex due to non-linearity, dynamic behaviour, and varying road conditions. To overcome these challenges. Dynamic temporal position observant graph neural network (DTPO-GNN) is formally presented as a novel architecture. This model incorporates positional awareness and spatial-temporal dependencies in traffic flow, using a controlled sample encoder-decoder architecture and two-way random walks. DTPO-GNN proves to be an efficient and accurate solution for large-scale road network traffic prediction [36].

The traffic graph convolutional recurrent neural network (T-GCRNN), proposed in [37], addresses large-scale traffic prediction by combining graph convolutional neural networks (GCNNs) to capture spatial relationships and recurrent neural networks (RNNs) to model temporal patterns in traffic data. Extensive experiments on real-world datasets demonstrate that T-GCRNN achieves superior performance, especially for long-term traffic forecasting. Similarly, the GCN-DHSTNet model effectively forecasts crowd flows, a task complicated by spatial-temporal dependencies and external influences such as weather conditions and the distribution of points of interest (POI). By integrating GCN with LSTM networks, GCN-DHSTNet leverages multiple temporal granularities and external features to deliver enhanced predictive accuracy [38].

Building on the importance of graph topology, advances in skeleton-based action recognition have led to the development of the ACC-GCN model, which introduces attentive channel-wise correlation graph convolution (ACC-GC). This method improves graph learning by capturing detailed topological relationships at the channel level, thereby enhancing correlation accuracy. The model achieves competitive benchmark results while reducing computational overhead [39].

To address the dynamic and complex spatio-temporal relationships in traffic forecasting, the dynamic multi-fusion graph network (DMFGNet) introduces an innovative strategy for dynamically capturing these correlations. Its core component, the spatio-temporal attention unit (STAU), optimizes neighbour aggregation weights to improve spatiotemporal integration. Validated on real-world datasets within a hybrid edge-cloud infrastructure, DMFGNet delivers exceptional performance, reduced latency, and efficient resource usage, highlighting the versatility and effectiveness of graph-based models across various applications [39].

3 Methodology

This research applied advanced deep-learning models to predict traffic flow at intersections. To tackle the complexities of traffic forecasting, we utilized well-established models such as gated recurrent units (GRU) [40], long short-term memory (LSTM) networks [41], and Convolutional Neural Networks (CNN) [42]. These models were chosen for their

effectiveness in capturing the temporal and spatial dynamics critical for traffic prediction. Given the spatiotemporal nature of traffic data, we experimented with various combinations, including LSTM-CNN, GRU-LSTM, and CNN-GRU. Our experiments revealed that the proposed Hybrid CNN-GRU-LSTM model achieved the highest accuracy.

The Hybrid CNN-GRU-LSTM model combines the strengths of CNN for spatial feature extraction and GRU-LSTM layers for modelling temporal dependencies. CNN efficiently captures the spatial correlations between road segments by analyzing vehicle flow patterns in adjacent areas, while the GRU and LSTM layers collaboratively learn both short-term fluctuations and long-term trends in traffic flow data. This hybrid approach is essential for accurate traffic forecasting, where both spatial proximity and temporal dynamics are crucial. We evaluated the performance of Hybrid CNN-GRU-LSTM across multiple datasets, including the PeMS dataset, England dataset, P/Castellano dataset, and Fedesoriano dataset. These datasets represent diverse traffic conditions, from urban roads to highways, enabling comprehensive testing of the model across different scenarios. The proposed model outperformed other combinations in terms of key metrics such as RMSE, MAE, and MAPE. The lower values of these metrics for Hybrid CNN-GRU-LSTM signify better predictive accuracy compared to existing models and combinations.

To assess the effectiveness of each model, we carefully performed data acquisition, preprocessing, feature engineering, model training, and evaluation. During the model assessment phase, we applied the aforementioned metrics to quantify the prediction accuracy. The results consistently showed that Hybrid CNN-GRU-LSTM produced superior outcomes in terms of both short-term and long-term traffic flow forecasts.

Once trained and tested, the model was deployed in a production environment to predict future traffic conditions. The model's ability to handle large datasets and complex spatial-temporal patterns makes it an invaluable tool for real-time traffic management. Figure 2 illustrates the detailed architecture of Hybrid CNN-GRU-LSTM, demonstrating how the CNN layer captures spatial dependencies and how GRU-LSTM layers jointly model temporal dependencies. Hybrid CNN-GRU-LSTM provides a robust solution for traffic prediction by integrating the advantages of CNN, GRU, and LSTM into a unified framework. Its application to various datasets showcases its adaptability and accuracy, significantly improving upon the predictive capabilities of existing models. The results from our experiments validate that Hybrid CNN-GRU-LSTM offers a reliable and effective approach to mitigating traffic congestion and enhancing transportation systems.

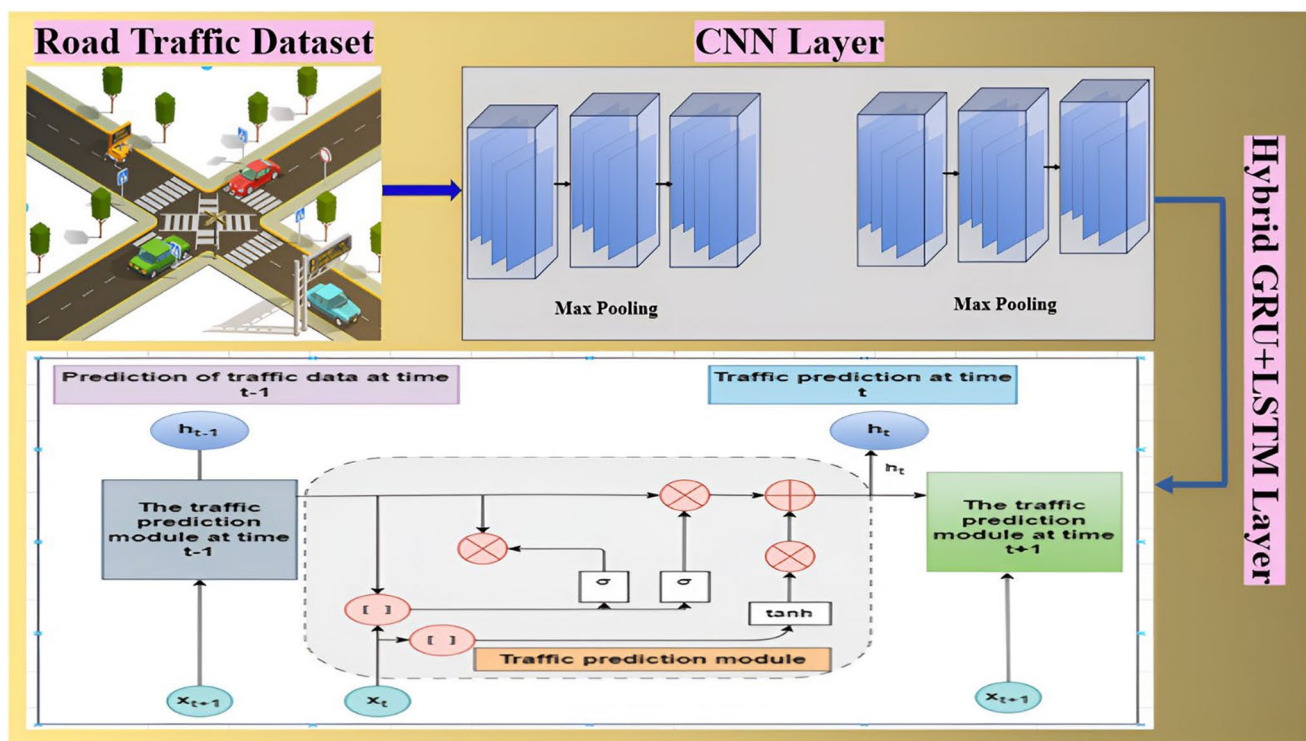


Fig. 2 Hybrid CNN-GRU-LSTM model

3.1 Hybrid CNN-GRU-LSTM architecture

The proposed methodology introduces a Hybrid CNN-GRU-LSTM architecture, as depicted in Fig. 2. This integrated model combines CNN, GRU, and LSTM layers to effectively capture both spatial and temporal dependencies inherent in traffic data, thereby enhancing prediction accuracy. The model's hyperparameters are detailed in Table 1. The architecture initiates with a 1D Convolutional Layer utilizing 32 filters with a kernel size of 3, performing one-dimensional convolution on the input data. The Rectified Linear Unit (ReLU) activation function is employed to introduce non-linearity, facilitating the extraction of spatial features from the data.

The CNN-GRU-LSTM hybrid model outperforms individual architectures due to its ability to combine the strengths of its components. The CNN layer is highly effective at extracting spatial features and identifying localized patterns, such as sudden variations or correlations within traffic sequences. The GRU and LSTM layers handle temporal dependencies, with GRU efficiently capturing short-term trends and LSTM excelling at learning long-term dependencies through its advanced gating mechanisms. This integration allows the model to represent both immediate and extended temporal contexts, improving its ability to capture periodic trends and sudden changes. Moreover, the hybrid structure reduces the risk of overfitting by balancing the simplicity of GRU with the robustness of LSTM. This comprehensive approach results in higher accuracy and better performance compared to standalone CNN-LSTM or GRU-LSTM models, making it particularly effective for complex predictive tasks.

In addition to its advantages, the model holds significant potential for practical applications in real-world traffic management, particularly in traffic signal control. By accurately predicting traffic flow at specific junctions, the model enables dynamic adjustments to signal timings based on anticipated congestion. This real-time adaptability optimizes traffic flow, reduces delays, and enhances overall traffic efficiency. Such capabilities are essential for the development of smart traffic systems, where data-driven decision-making can lead to more responsive and effective transportation networks. Through these practical applications, the model bridges the gap between theoretical potential and real-world use, contributing to smarter and more efficient urban traffic management.

Subsequent to the Conv1D layer, a MaxPooling1D layer is incorporated to reduce data dimensionality. Operating with a pool size of 2, this layer identifies and extracts the most significant features from the preceding convolutional layer. This sequential integration of Conv1D and MaxPooling1D layers optimally contributes to the model's ability to discern spatial intricacies and prioritize essential information, thereby bolstering its efficacy in making accurate predictions within the context of traffic data analysis. A Dropout layer with a dropout rate of 0.5 is used to reduce overfitting. Subsequently, two Dense layers are added, with the first one having 100 units and ReLU activation. The final Dense layer with one unit is utilized for the output prediction.

$$\text{conv}(x) = \sigma(\omega * x + b) \quad (1)$$

Equation (1) represents a simple mathematical convolutional neural network layer model. Where $\text{Conv}(x)$ represents the output of the convolutional layer when given an input 'x'. This output is a set of feature maps or activations that capture certain patterns or features present in the input. And 'w' and 'b' are the parameters of the convolutional layer where 'w'

Table 1 Hyper parameter of hybrid CNN-GRU-LSTM

Model	Hyperparameter	Value
Hybrid CNN-GRU-LSTM	Model	Sequential
	Batch_size	150
	Filter	32
	Kernal_size	3
	Patience	10
	Learning rate	0.001
	Number of epochs	100
	Number of hidden layers	3
	Hidden unit	256
	Dropout	0.5
	Activation function	relu, tanh
	Output unit	1
	Output type	Single label
	Optimizer	Adam
	Loss function	Mean squared error

represents a set of learnable filters, also known as weights, that are convolved with the input 'x' to produce the feature maps, and 'b' represents a bias term that is added to each feature map. 'σ' represents an activation function, such as the sigmoid or ReLU function, that applies a non-linear transformation to the output of the convolutional layer. Equation (1) represents a simple representation of processing road data with spatial structure using convolutional neural networks.

Subsequent to the Convolutional layers, GRU Layer Gated Recurrent Unit (GRU) layer is incorporated to capture the temporal dependencies in the data. The GRU layer has 100 units, which represent memory cells capable of retaining information over time. By returning sequences, this layer can pass information to subsequent layers. Dropout Layer prevents overfitting, a Dropout layer is added with a dropout rate of 0.5. Dropout randomly sets a fraction of input units to 0 during training, reducing the model's reliance on any single feature and improving its generalization.

$$h_t^{(i)} = GRU(x_t, h_{t-1}^{(i)}) \quad (2)$$

Equation (2) represents a single time step in a recurrent layer using a gated recurrent unit (GRU). $h_t^{(i)}$ hidden state vector at time step 't' for the i th neuron (or unit) in the recurrent layer. This vector summarizes the information learned by the neuron from the input sequence up to that point. GRU is an advanced techniques of RNN cell that incorporates gating mechanisms to help address the vanishing gradient problem. The GRU function takes two inputs: 'i' index of the current neuron being processed. ' x_t ' input vector at time step 't' that the current neuron is processing. ' $h_{t-1}^{(i)}$ ' hidden state vector from the previous time step $t - 1$ that is used as additional information to help update the current hidden state vector ' $h_t^{(i)}$ '.

In the model's final stage, LSTM layers are added. LSTM, a recurrent neural network, is tailored for handling time series data. It employs specialized units with input, output, and forget gates and an internal cell state to retain and utilize information over long sequences. This architecture helps address the vanishing gradient problem commonly encountered in standard RNNs. LSTM's ability to selectively retain or discard information enables it to effectively capture temporal dependencies, making it ideal for tasks like time series forecasting and natural language processing model for time series prediction, specifically for traffic prediction. The LSTM model is built using TensorFlow Keras' Sequential API. It includes two LSTM layers, each with 100 units, followed by Dropout layers with a rate of 0.5 to minimize overfitting. The model is compiled with the Adam optimizer and the MSE loss function.

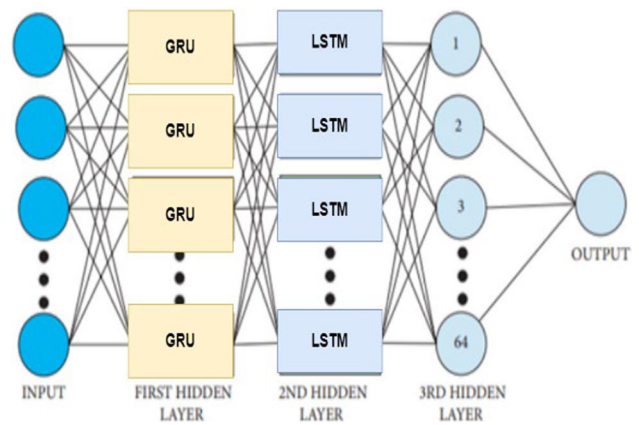
$$h_t^{(i)} = LSTM(x_t, h_{t-1}^{(i)}, c_{t-1}^{(i)}) \quad (3)$$

In LSTM h_t represent as hidden state at time 't', which is a vector that summarizes the relevant information from previous time steps and the current input ' c_t ' is the cell state, which is also a vector that stores information over time.

3.2 GRU-LSTM and CNN-LSTM architecture

The GRU-LSTM model is implemented using the TensorFlow library with Keras as the high-level API. GRU-LSTM model represent in Fig. 3, model architecture starts with a GRU layer, which processes sequential data and captures temporal dependencies effectively. The GRU layer is followed by a dropout layer to prevent overfitting during training. The dropout layer helps in regularization and improves the model's generalization ability. Also, we include an LSTM layer, which further refines the sequential information and considers long-term dependencies within the traffic data. The LSTM layer also helps in mitigating the vanishing gradient problem, which can occur in deep neural networks during backpropagation. A dense layer with a single output unit is added to complete the model architecture, enabling the model to predict the number of vehicles as a regression task. The model is trained using the mean squared error loss function and the Adam optimizer, which efficiently updates the model's weights during training.

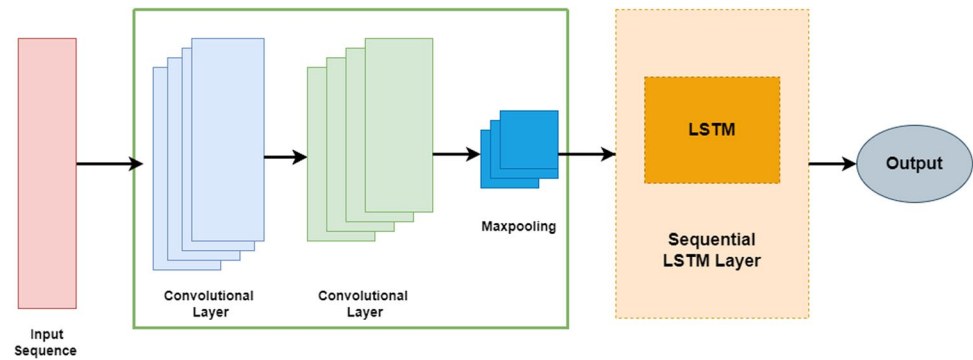
We implement an early stopping mechanism with a minimum delta of 0.001 and a patience of 100 epochs to prevent overfitting and facilitate early convergence. This ensures that the training halts if no significant improvement in validation loss is observed. The hyperparameters for the GRU-LSTM model are detailed in Table 2, whereas same for CNN-LSTM are shown in Table 3. The CNN-LSTM model that combines Convolutional Neural Network (CNN) layers with Long Short-Term Memory (LSTM) layers, is illustrated in Fig. 4. This architecture is tailored to capture spatial patterns using CNN layers and temporal patterns with LSTM layers, making it well-suited for analyzing traffic time series data with spatial characteristics. The model includes a 1D Convolutional layer with 64 filters and a kernel size of 3, followed by a MaxPooling1D layer with a pool size of 2 to reduce dimensionality. Two LSTM layers, each with 100 units, are then added, followed by Dropout layers with a 0.5 rate to combat overfitting. The output prediction is generated by a Dense layer with a single unit. The model is compiled with the Adam optimizer and the mean squared error (MSE) loss function. Training is conducted over

Fig. 3 GRU-LSTM architecture**Table 2** Hyper parameter of hybrid GRU-LSTM

Model	Hyperparameters	Value
GRU-LSTM	Model	Sequential
	Batch_size	150
	Number of GRU layers	2
	Number of LSTM layers	1
	GRU unit	256
	LSTM unit	256
	Patience	10
	Learning rate	0.001
	Number of epochs	100
	Dropout	0.5
	Activation function	Relu, tanh
	Output unit	1
	Output type	Single label
	Optimizer	Adam
	Loss function	Mean squared error

Table 3 Hyper parameter of hybrid CNN-LSTM

Model	Hyperparameters	Value
CNN-LSTM	Model	Sequential
	Batch_size	150
	Number of CNN layers	2
	Filter	32
	Kernal_size	3
	Number of LSTM layers	1
	LSTM unit	256
	Patience	10
	Learning rate	0.001
	Number of epochs	100
	Dropout	0.5
	Activation function	Relu, tanh
	Output unit	1
	Output type	Single label
	Optimizer	Adam
	Loss function	Mean squared error

Fig. 4 CNN-LSTM architecture

100 epochs with a batch size of 150, and early stopping is utilized as a callback with a minimum delta of 0.001 and a patience of 10 epochs to avoid overfitting and retain the optimal weights.

4 Dataset description

This research employs multiple publicly available traffic datasets within our proposed Hybrid CNN-GRU-LSTM approach. These datasets encompass diverse features and attributes, contributing to the comprehensive analysis conducted in this research.

4.1 England dataset

The analysis of the traffic dataset of England is gathered from the Highways England Site [43], capturing traffic flow dynamics from January 2020 to November 2020. To explain traffic patterns, particular emphasis is placed on peak hours occurring on weekdays, specifically from 8 to 9 AM. This dataset comprises a comprehensive set of observations spanning 204 days. The dataset contains the parameters like Day Type ID, Carriageway flow, Speed Value and the type of vehicle variant in size for every 15 min interval. Figure 5 represents the aggregated values in a day for each parameter. Before employing time series forecasting models, the dataset undergoes preprocessing procedures, including normalization through the Z-Score Normalization technique. This normalization step is crucial for ensuring the data is appropriately standardized, facilitating effective analysis and modeling processes.

4.2 PeMS dataset

Real traffic data was sourced from the Caltrans via the PeMS [44] platform, a traffic monitoring system that provides data from California's road network. The dataset includes traffic flow, mean speed, and traffic density measurements obtained from the I5 freeway, chosen due to its extensive coverage and numerous traffic detectors. Three sections of the I5 freeway were selected, with traffic data collected at 5-minute intervals for 2015 and 2016, totaling 600,000 samples. PeMS traffic flow values, initially recorded every 5 min, are converted to vehicles per hour per lane to standardize the input for neural networks. This conversion abstracts the networks from the specific lane configurations of each road section. The

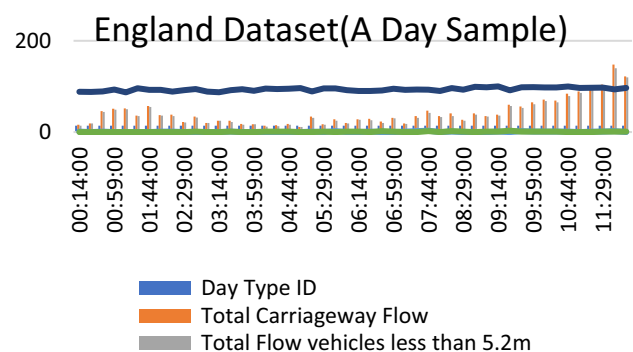
Fig. 5 England dataset exploratory analysis

Fig. 6 PeMS dataset exploratory analysis

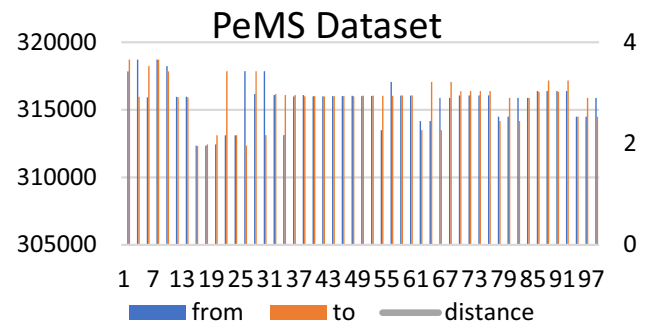
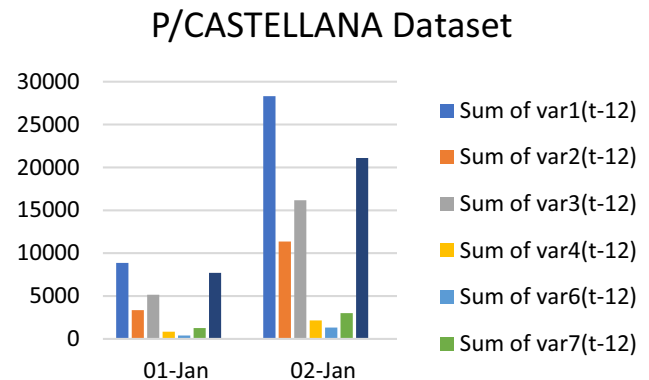


Fig. 7 P/Castellana dataset exploratory analysis



features like distance between the nodes and the source-destination along with traffic flow information are provided in the database as shown in Fig. 6.

4.3 P/Castellana dataset

In this dataset [45], we identify the predictor variables and the target variable while also specifying the data sources utilized for our analyses. The target variable represents the hourly count of vehicles passing through a designated traffic station in a specific direction. To evaluate our model, we selected four key traffic stations in Madrid, each representing areas with notable traffic volumes and unique features. Among these stations, three are situated in the city's core districts, serving as key nodes in the urban traffic network. Additionally, we included one station located at a major entry/exit point of the city. Specifically, Paseo de la Castellana, one of our selected stations, gathers data in the North–South direction, serving as a significant thoroughfare traversing Madrid from its downtown area towards the North. As a predictor variable, we have also considered the hour in the timestep. For each prediction, the input data is offered in four different hours, which is aggregated day-wise, as shown in Fig. 7.

4.4 Fedesoriano dataset

The dataset [46] contains date-time information and vehicle count data at a traffic junction. The data spans from November 1, 2015, to June 30, 2016. To improve the data, additional time-based features were derived, including year, month, day, hour, and day of the week. These temporal features are subsequently employed for visual analysis, exploring the relationships between time components and the 'Vehicles' variable. Figure 8 presents visualizations with time-related features on the x-axis (e.g., year, month), while the 'Vehicles' count is plotted on the y-axis. Different lines or colors represent distinct 'Junction' values. These plots facilitate the identification of temporal patterns in vehicle counts at various junctions, providing insights into trends and variations in traffic flow across different times and days. The data collected through traffic sensors supports a detailed analysis of traffic patterns, offering a deeper understanding of vehicle movement dynamics at different times of day and week.

After preprocessing, the dataset is divided into training, validation, and testing subsets. The hybrid CNN-GRU-LSTM model is trained on the training set, optimized with the validation set to adjust hyperparameters, and evaluated using the testing set to measure its performance.

Every intersection of the junction has been seeing a general increasing trend over the last year. As we have established before, the fourth junction only contains a small amount of data, and that data would not go back more than a year. During the month of June, there is a noticeable increase in traffic at the first and second crossroads. Our presumption is that this is because of the summer holidays and the activities that surround them. Monthly, there is a decent consistency in all the data over all dates. We can observe that activity is at its highest throughout the peak working hours of the day and that it drops down significantly during the night time hours. This is in line with what was anticipated. When considering weekly trends, Sundays often have less congestion than other days of the week since fewer automobiles, buses and bikes are on the road. However, the volume of traffic is rather constant throughout the weekdays. In this research paper, we observed an upward trend in the yearly data for all four junctions, as mentioned earlier, the first and second junctions experienced an arrival of traffic around June, which could be attributed to summer break and related activities. Additionally, we found a consistent pattern in the data on a monthly basis, regardless of the date. Daily, traffic peaks occur during morning and evening hours, followed by a decline during the night, which is consistent with our expectations. Finally, we analyzed the weekly traffic patterns and found that Sundays experience smoother traffic with fewer vehicles on the road, while traffic remains steady from Monday to Friday.

5 Experimental setup

The proposed traffic prediction models run on a Windows operating system, which is configured with a Core i5 processor and 16 GB of primary memory.

5.1 Performance and parameter

To assess the effectiveness of our proposed hybrid model, we calculate RMSE, MAE, MAPE, and MSE. RMSE is a metric for quantifying the disparity between actual and predicted traffic congestion. Equation (5) is applied to evaluate RMSE, continuously measuring error rates. Additionally, MAE and MAPE are computed using Eqs. (6) and (7), respectively, elucidating the model's predictive accuracy. Furthermore, MSE is calculated using Eq. (8), providing insight into the average squared differences between predicted and actual values.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (predict - actual)^2} \quad (5)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |predict - actual| \quad (6)$$

$$MAPE = \frac{1}{n} \sum \left| \frac{actual - predict}{actual} \right| \quad (7)$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (predict - actual)^2 \quad (8)$$

5.2 Hybrid CNN-GRU-LSTM algorithm

To instantiate this hybrid model, the Python programming language is connected, in conjunction with the Colab notebook, to create an integrated development environment. Keras and TensorFlow, as deep learning libraries, are imported to facilitate the formulation of an artificial neural network. The architecture of this neural network is characterized by three distinct layers.

Firstly, the input layer serves as the initial entry point for the model, accommodating the traffic data alongside their associated dates, which correspond to the number of vehicles recorded. Subsequently, the model employs a CNN layer, designed to extract spatial features from the input data. This layer serves as a pivotal component in spatial feature

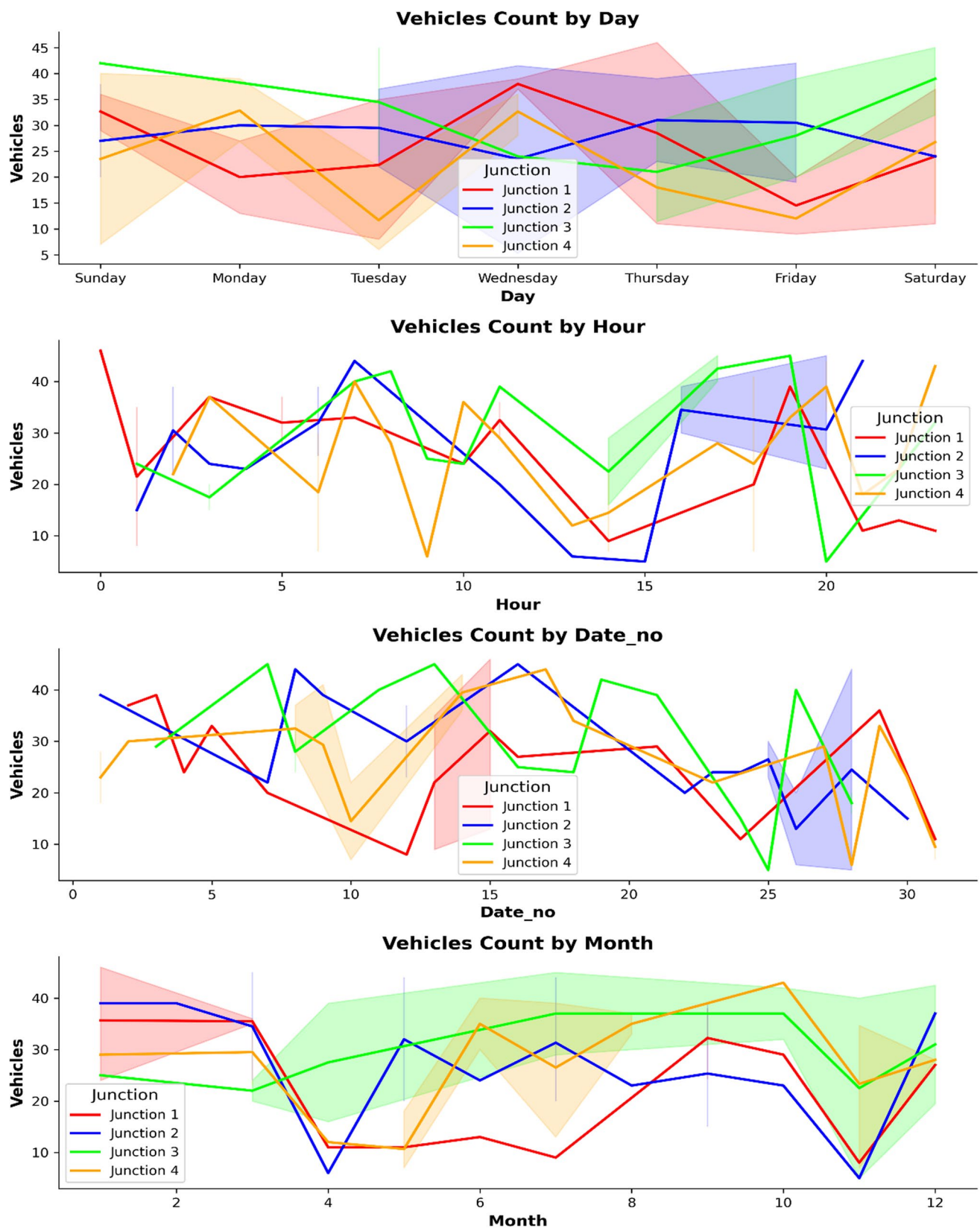


Fig. 8 Day, month, date number and hour pre-processed data

recognition within the traffic data. The output from the CNN layer is seamlessly propagated to the hybrid model, combining GRU and LSTM layers, collectively referred to as the GRU-LSTM hybrid model. This amalgamated architecture is equipped with sigmoid and hyperbolic tangent (tanh) activation functions, enabling the sophisticated processing of the input data by capturing temporal dependencies, which are instrumental in making accurate predictions. To visualize and compare the true values with the predicted values, the *"PredictionsPlot"* function is invoked. In the visual representation, the true values are distinctly labeled as *'True Value'* on the graph. The predicted values, which are generated by the hybrid GRU-LSTM model for corresponding inputs in the test dataset, are obtained by utilising the *"model.predict(X_test)"* function within the Hybrid CNN-GRU-LSTM model function. These predicted values are stored in the variable named *'predicted'* and are presented on the graph under the label *'Predicted Values'*. Evaluating the model's performance is imperative, which is accomplished by calculating RMSE. The *"RMSE_Value"* function is responsible for this computation, accepting the true values and predicted values from the test dataset as inputs. RMSE, a metric measuring the disparity between the true and predicted values, is returned by the *"RMSE_Value"* function, with lower RMSE values indicating superior predictive performance. In addition to RMSE, MAE and MAPE are critical performance metrics. These are calculated through the *"MAE_Value"* and *"MAPE_Value"* functions, respectively. Both functions take the true values and predicted values from the test dataset as input and return the corresponding MAE and MAPE values. These metrics offer further insights into the model's effectiveness in predicting traffic congestion, with lower MAE and MAPE values signifying enhanced predictive accuracy and predicted values.

6 Result and analysis

This research introduces a hybrid deep-learning model to address traffic congestion prediction, which optimizes through the minimization of a loss function using backpropagation. This loss function encompasses key metrics such as MSE, RMSE, MAPE, and MAE to quantitatively measure the disparities between predicted and actual values. Our proposed model employs parameter tuning techniques, including adjusting hyperparameters such as learning rate, batch size, and the number of training epochs, executed using a validation set to enhance performance. After training and fine-tuning, the model undergoes comprehensive evaluation using various performance metrics, including MAPE, RMSE, and MAE. Our hybrid CNN-LSTM-GRU model stands out as an efficient method for traffic prediction, benefiting from its fusion of convolutional and recurrent layers. This unique architecture allows it to capture spatial and temporal dependencies within the input data, leading to highly accurate predictions. The superiority of the Hybrid CNN-GRU-LSTM model can be attributed to its ability to exploit the complementary strengths of CNNs for spatial feature extraction and GRUs/LSTMs for temporal modelling. By combining these architectures, the model can effectively capture the intricate spatial-temporal relationships inherent in traffic data, resulting in improved learning and prediction performance. Additionally, the gated mechanisms in GRUs and LSTMs enable them to mitigate the vanishing gradient problem and handle long-range dependencies in sequential data more effectively. Consequently, the model consistently achieves lower error metrics, such as RMSE and MAE, than other models, making it the preferred choice for traffic prediction tasks due to its enhanced accuracy and robust performance.

6.1 Comparative analysis using England dataset

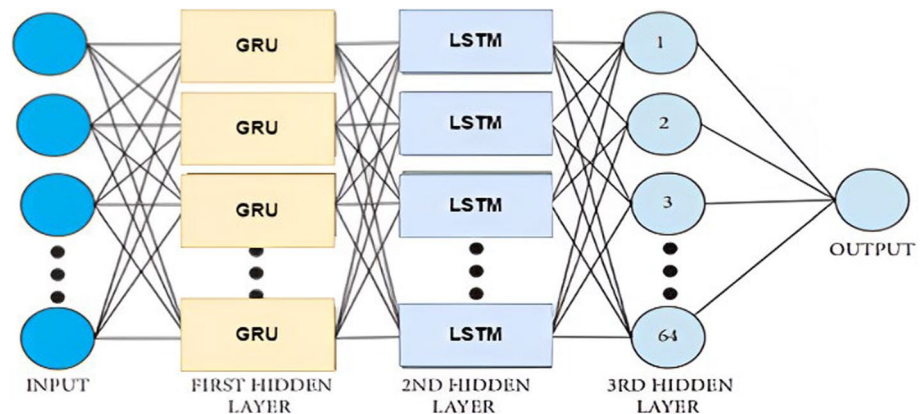
The dataset used in this study contains detailed traffic flow data collected over a period spanning from January 2020 to November 2020, specifically focusing on peak traffic hours during weekdays from 8 to 9 AM. This dataset encompasses observations gathered over a total of 204 days, providing a comprehensive representation of traffic patterns in urban areas. Before applying this dataset to time series forecasting tasks, it is preprocessed to ensure its suitability for analysis. During preprocessing, data cleaning, normalization, and feature engineering techniques may be applied to enhance the quality and usability of the dataset.

The predictive model utilized in this study is a Hybrid CNN-GRU-LSTM model, which integrates CNN, GRU and LSTM. This model architecture is trained using 163 days (approximately 80%) of the dataset, while the remaining 41 days (approximately 20%) are reserved for testing purposes. The training phase involves optimizing the model parameters using a training dataset to minimize prediction errors and enhance accuracy.

A comparative result analysis of the model's performance is conducted using key metrics such as MSE, MAPE, and MAE. The results of this analysis are presented in Table 4, which provides a side-by-side comparison of the performance of the Hybrid CNN-GRU-LSTM model with other baseline models, including CNN, LSTM, GRU, and Hybrid CNN-LSTM.

Table 4 Comparative analysis using England dataset with proposed approach

Error matrix	CNN	LSTM	GRU	CNN-LSTM [47]	GRU-LSTM	Hybrid CNN-GRU-LSTM [Proposed]
MSE	0.3096	0.2123	0.1451	0.0568	0.0813	0.0431
MAE	0.4011	0.3379	0.0312	0.9878	0.6741	0.8165
MAPE %	28	26.17	11.43	11.07	0.5349	7.86

Fig. 9 Actual and predicted value at each 5-min interval using Hybrid CNN-GRU-LSTM England dataset)

Hybrid CNN-GRU-LSTM model emerges as the top performer across all metrics, showcasing its superiority in forecasting traffic flow. With significantly lower MSE, MAE, and MAPE values of 0.0431, 0.8165, and 7.86% respectively, the hybrid model effectively minimizes prediction errors and enhances accuracy compared to its counterparts. This exceptional performance can be attributed to the synergistic integration of CNN, GRU, and LSTM layers, which enable the model to capture both spatial and temporal information inherent in the dataset.

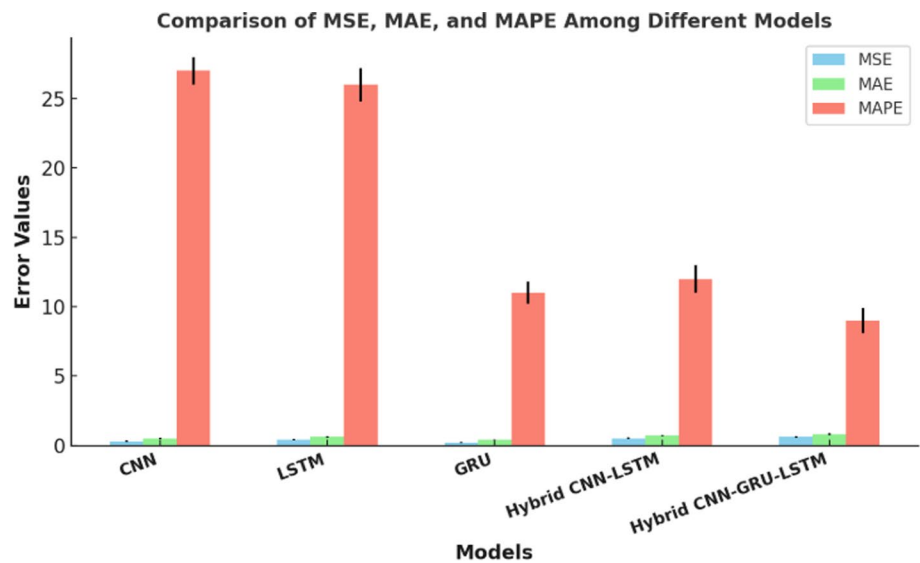
The comparative analysis underscores the potential of the proposed Hybrid CNN-GRU-LSTM model for achieving superior performance in predictive analytics applications, particularly in the context of traffic forecasting. By effectively leveraging advanced deep learning techniques and integrating spatial and temporal information, this model offers valuable insights for optimizing traffic management and urban planning efforts. Figures 9 and 10 present the results for this dataset.

The plot in Fig. 9 depicts the actual and predicted values for a Hybrid CNN-GRU-LSTM model illustrated in Fig. 9, a time series comparison between actual value and predicted values over a span of 55 min. The x-axis represents time in minutes, while the y-axis shows the value range from 0 to 1. Actual values are depicted with a red line and markers, whereas predicted values are shown with a blue line and markers. The data points reveal that the predicted values generally follow the trend of the actual values, although there are notable deviations at certain points, such as around 20, 30, and 40 min, where the predictions either overshoot or undershoot the actual values. This visual representation helps assess the model's performance and identify areas for improvement in prediction accuracy.

6.2 Comparative analysis using PeMS dataset

The performance evaluation of our proposed Hybrid CNN-GRU-LSTM model on the PeMS dataset provides its better predictive capabilities than other models, as evidenced by the significantly lower RMSE and MAE values. Specifically, our model achieves an RMSE of 16.05 and an MAE of 9.76, outperforming CNN, LSTM, GRU, and even the CNN-LSTM hybrid model. This substantial reduction in both RMSE and MAE underscores the effectiveness of our model in accurately capturing the intricate patterns and temporal dependencies present in the data. By integrating convolutional, gated recurrent, and long short-term memory layers, our Hybrid CNN-GRU-LSTM model demonstrates a remarkable ability to discern complex relationships within the dataset, leading to more precise and reliable predictions.

Similarly, when evaluating forecasting performance on the Paseo de la Castellana dataset, our Hybrid CNN-GRU-LSTM model consistently outperforms other models across all prediction horizons, as indicated by lower RMSE and MAE values. Across different forecast intervals, the RMSE values range from 219.13 to 284.27, while MAE values range from 142.65

Fig. 10 Comparative analysis of England dataset

to 179.54. This consistent superiority underscores the efficacy of integrating CNN, GRU, and LSTM layers in capturing both spatial and temporal dependencies within the dataset. The Hybrid CNN-GRU-LSTM model's ability to effectively leverage these diverse architectures enables it to extract meaningful insights from the data and produce highly accurate forecasts. Consequently, our model emerges as the optimal choice for precise and reliable forecasting tasks on the Paseo de la Castellana dataset, offering valuable insights for various applications in traffic management and urban planning.

Table 5 and Fig. 11 present the comparative analysis of the PeMS dataset. These error values are the lowest for the proposed model. Further, it can be noted that the RMSE and MAE values for the proposed model are the lowest (best); moreover, the difference between the RMSE and MAE of the proposed model for the PEMS dataset is not very high, implying the absence of huge individual error values.

6.3 Comparative analysis using P/Castellana dataset

The dataset utilized in our experiments comprises traffic flow data gathered from four traffic monitoring stations across Madrid, specifically chosen for their high traffic volumes and distinct characteristics. One of the stations, located on Paseo de la Castellana (P/Castellana), captures traffic data along the North–South axis. As a major thoroughfare extending from downtown Madrid to the northern region, Paseo de la Castellana serves as a critical traffic corridor within the city. The primary variable of interest in our analysis is the hourly vehicle count at each monitoring station, offering crucial insights into the traffic patterns and dynamics of various locations within Madrid.

In this dataset, we implemented experiments to validate the efficiency of our proposed Hybrid CNN-GRU-LSTM model for traffic prediction. The comparison presented in the tables provides the better performance of various models for traffic prediction across different time horizons. Traditional models like CNN and LSTM demonstrate moderate accuracy,

Table 5 Comparative analysis using the *PeMS* dataset using the proposed approach

Prediction model	RMSE	MAE
CNN	19.73	14.63
LSTM	19.37	14.27
GRU	18.11	15.87
GRU-LSTM	19.63	17.04
CNN-LSTM [48]	16.78	11.57
STwave [49]	26.50	14.93
STGODE [49]	27.30	15.71
AGCRN [49]	27.19	15.24
LSGCN [49]	28.41	16.76
HYBRID CNN-GRU-LSTM (proposed)	16.05	9.76

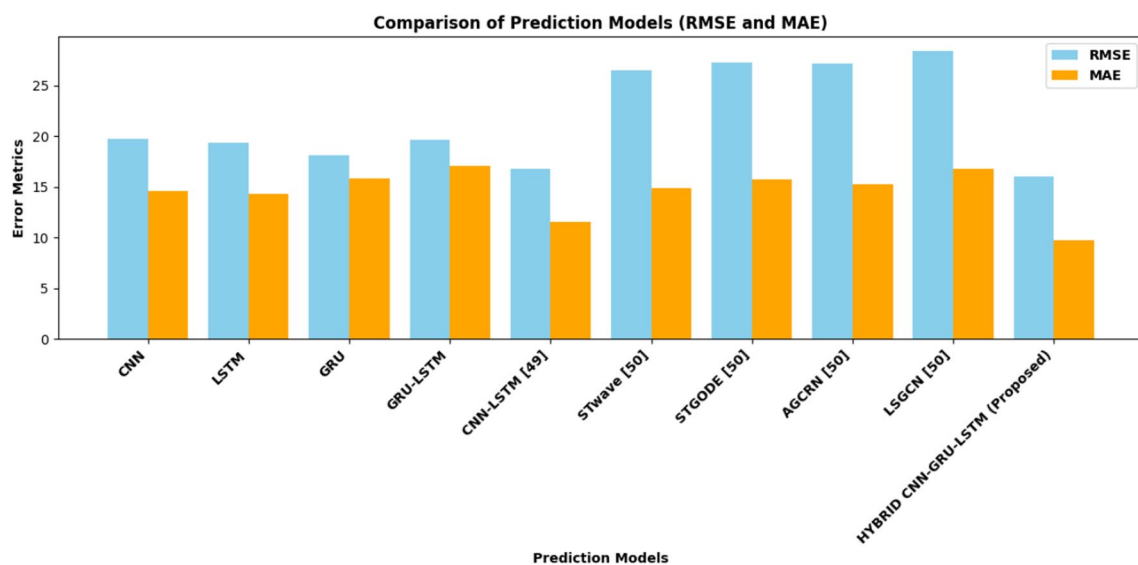


Fig. 11 Comparative analysis of PEMS dataset

with relatively higher RMSE and MAE values, indicating their limitations in capturing the complexities of traffic patterns. However, hybrid models that combine CNN with LSTM, BiLSTM, or GRU layers, such as CNN-LSTM, CNN-BiLSTM, and Hybrid CNN-GRU-LSTM, show improved predictive capabilities, reflected in lower error values. Among these hybrid models, Hybrid CNN-GRU-LSTM consistently stands out as the top performer, consistently achieving the less RMSE and MAE values across all prediction horizons. This superiority underscores the effectiveness of integrating convolutional, recurrent, and gated recurrent neural network architectures for traffic prediction tasks. These findings highlight the significance of hybrid models in enhancing prediction accuracy, which is crucial for making informed decisions in traffic management and urban planning contexts. By leveraging the strengths of different neural network architectures, hybrid models offer a promising approach to address the challenges of traffic prediction and improve overall system efficiency. The comparative results are shown in Tables 6 and 7.

Tables 6 and 7 present the comparative analysis of P/Castellano dataset. These error values are the lowest for the proposed model. Further, it can be noted that the RMSE and MAE values for the proposed model are the lowest (best); moreover, the difference between the RMSE and MAE of the proposed model for the P/Castellano dataset is also lowest for our model as compared to other models, implying the absence of huge individual error values.

6.4 Comparative analysis using Fedessoriano dataset

The dataset utilized in this research encompasses comprehensive information regarding date-time and vehicle counts at various junctions, collected meticulously from November 1, 2015, to June 30, 2016. To enrich the dataset and unlock its full potential, additional features are engineered based on date and time parameters. These enhancements include the extraction of various time-related components such as year, month, day, hour, and day of the week. By incorporating these supplementary attributes, we enable a more profound exploration of temporal patterns and their intricate relationships with traffic congestion dynamics. Within our model framework, Junctions J1 and J2 exhibit coherent traffic patterns, while Junction J3 demonstrates a more uniform distribution of vehicles with sporadic spikes. Conversely, Junction J4 displays a more uniform growth trajectory interspersed with intermittent spikes. Visual representations of these traffic patterns are depicted in Fig. 12 (a), (b), (c) and (d), where the X-axis delineates data attributes, and the Y-axis denotes the vehicle count passing through each junction. Each junction is represented by two distinct lines: one reflecting predicted values and the other portraying original values, thereby aiding in traffic congestion prediction.

An intriguing observation stemming from the dataset pertains to the collection of vehicle counts at Junction 4 over a comparatively shorter timeframe. This aspect, noted by some researchers, prompts a nuanced approach in our study. Consequently, our analysis is bifurcated into two segments: the first segment scrutinizes the RMSE and MAE values across all junctions, depicted in Fig. 13a, b, while the second segment zeroes in on RMSE and MAPE comparisons for three specific junctions shown in Fig. 14a–c. Our experimental outcomes are juxtaposed against established methodologies,

Table 6 Comparative analysis using (*P/Castellana*) dataset using the proposed approach with RMSE value

Model	RMSE			
	12 h	24 h	48 h	72 h
CNN	458.45	354.49	429.96	443.11
LSTM	257.14	256.76	294.71	320.66
CNN-LSTM	249.14	236.76	287.29	299.21
GRU-LSTM	265.18	298.51	278.65	298.70
CNN-BiLSTM [50]	224.46	241.47	277.24	291.11
HYBRID CNN-GRU-LSTM	219.13	239.53	281.14	284.27

Table 7 Comparative analysis using (*P/Castellana*) dataset using proposed approach using MAE value

Model	MAE			
	12 h	24 h	48 h	72 h
CNN	353.44	269.79	326.07	338.02
LSTM	157.61	153.76	187.74	197.81
CNN-LSTM	151.57	141.55	178.63	185.65
GRU-LSTM	167.32	189.11	176.43	189.95
CNN-BiLSTM [50]	144.22	136.51	168.92	183.95
HYBRID CNN-GRU-LSTM	142.65	131.98	159.13	179.54

showcasing a substantial enhancement as our proposed hybrid model consistently surpasses existing approaches by an impressive margin of 30–35%. Quantitative assessments of our model's predictive accuracy are gauged through RMSE values for each junction—Junction 1 (0.2391), Junction 2 (0.4740), Junction 3 (0.5775), and Junction 4 (1.0096), presented in Table 8. Additionally, complementary metrics such as MSE and MAPE are computed to provide a holistic evaluation, with lower MAPE values signifying superior model performance. These are presented in Table 9. These error values are the lowest for the proposed model. Further, it can be noted that the RMSE and MAE values for the proposed model are the lowest (best), thus showing its applicability in real-world traffic management scenarios.

Further, it can be noted from Tables 8 and 9 that the RMSE and MAE values for the proposed model are the lowest (best) for all the junctions; moreover, the difference between the RMSE and MAE of the proposed model for the dataset is not very high implying the absence of huge individual error values.

7 Conclusion and future scope

This research introduces a groundbreaking solution to the critical issue of traffic prediction in several datasets, each of which has different attributes like peak hours and traffic at junctions, where congestion significantly impacts transportation efficiency and urban life. The research demonstrates a substantial improvement in prediction accuracy by developing a novel Hybrid CNN-GRU-LSTM model, which combines GRU, LSTM, and CNN deep learning techniques to capture both temporal and spatial aspects of traffic patterns. Testing against public traffic data reveals up to a 30–35% reduction in error values compared to existing models, showcasing the model's superior performance. The uniqueness of our work lies in applying the CNN layer for spatial feature mining and multi-layer GRU-LSTM for better accuracy in traffic flow prediction. This research offers a compelling path forward for transportation management, enabling proactive measures and enhancing transportation network efficiency to alleviate congestion-related challenges, reduce economic losses, and mitigate environmental pollution in cities worldwide.

In the future, the direction of the research may emphasize refining the model's performance and robustness by experimenting with various configurations and techniques, such as regularization methods, learning rate schedules, and advanced optimization algorithms [53], as well as working with video-based traffic datasets. These efforts will contribute to the development of more effective and efficient traffic prediction solutions that can better address the dynamic nature of urban traffic environments.

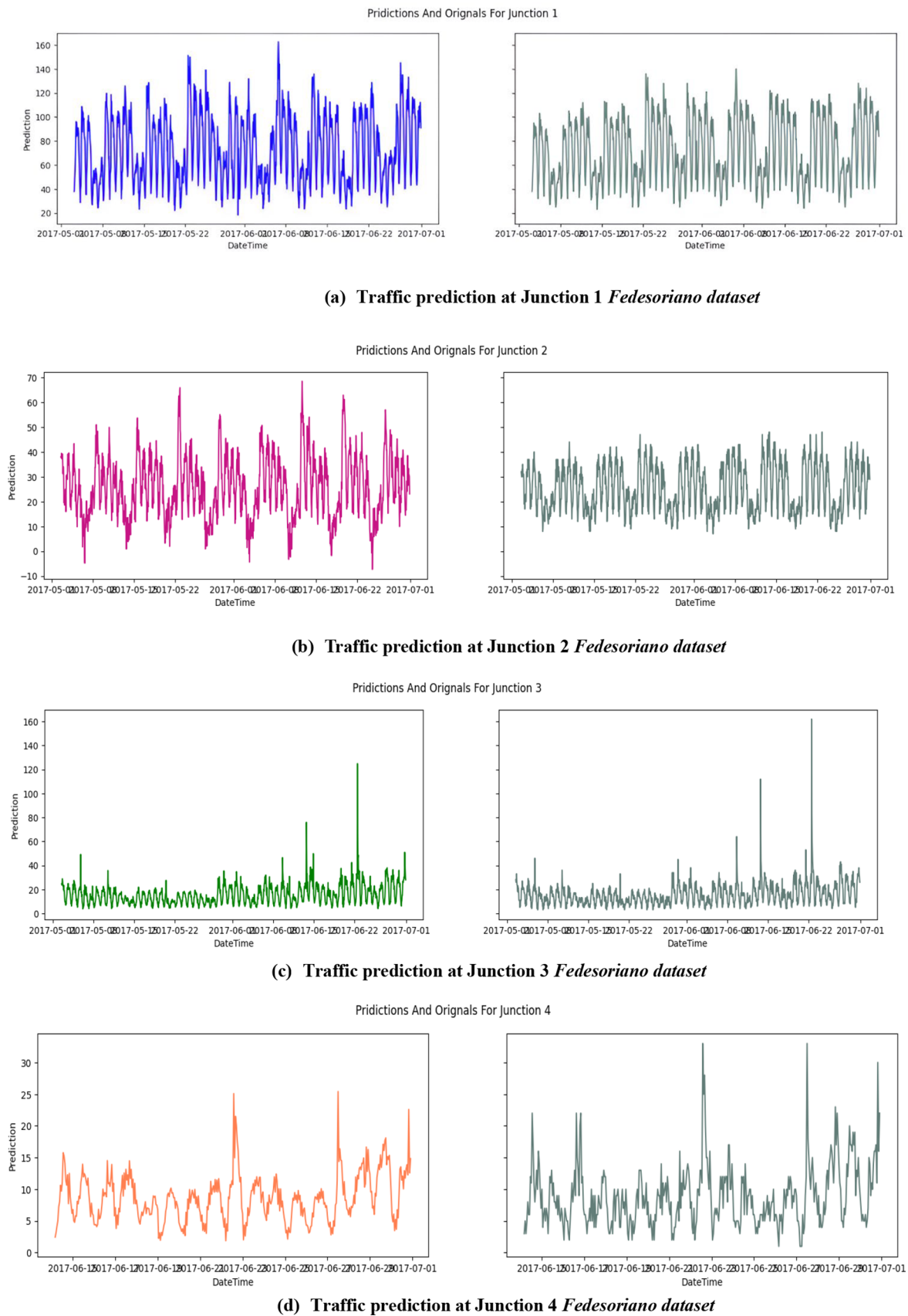
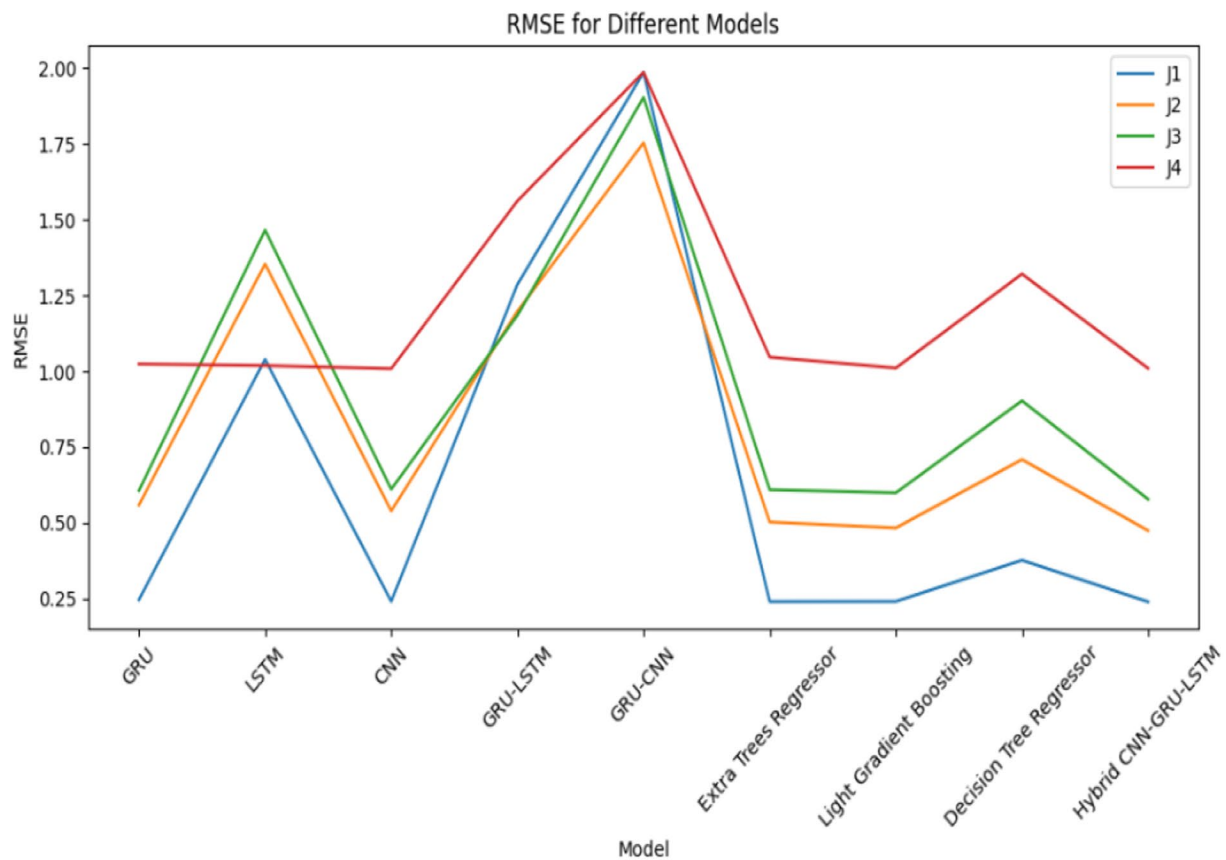
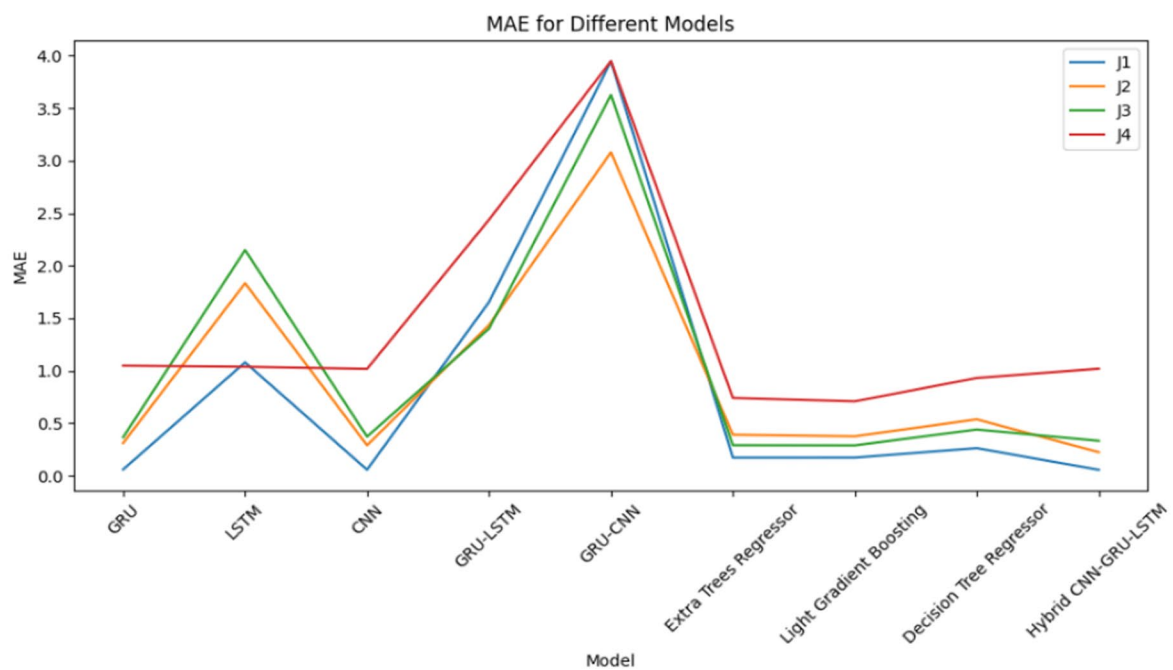


Fig. 12 **a** Traffic prediction at Junction 1 Fedesoriano dataset. **b** Traffic prediction at Junction 2 Fedesoriano dataset. **c** Traffic prediction at Junction 3 Fedesoriano dataset. **d** Traffic prediction at Junction 4 Fedesoriano dataset

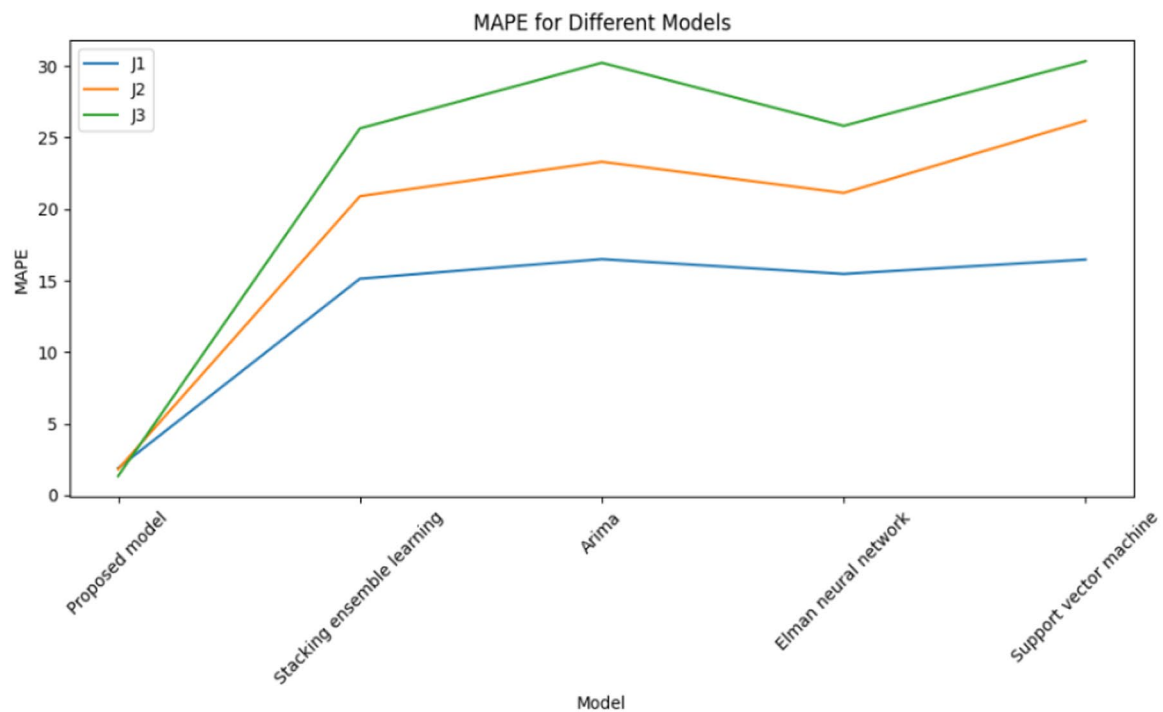


(a) RMSE Comparison with different models using *Fedesoriano* dataset

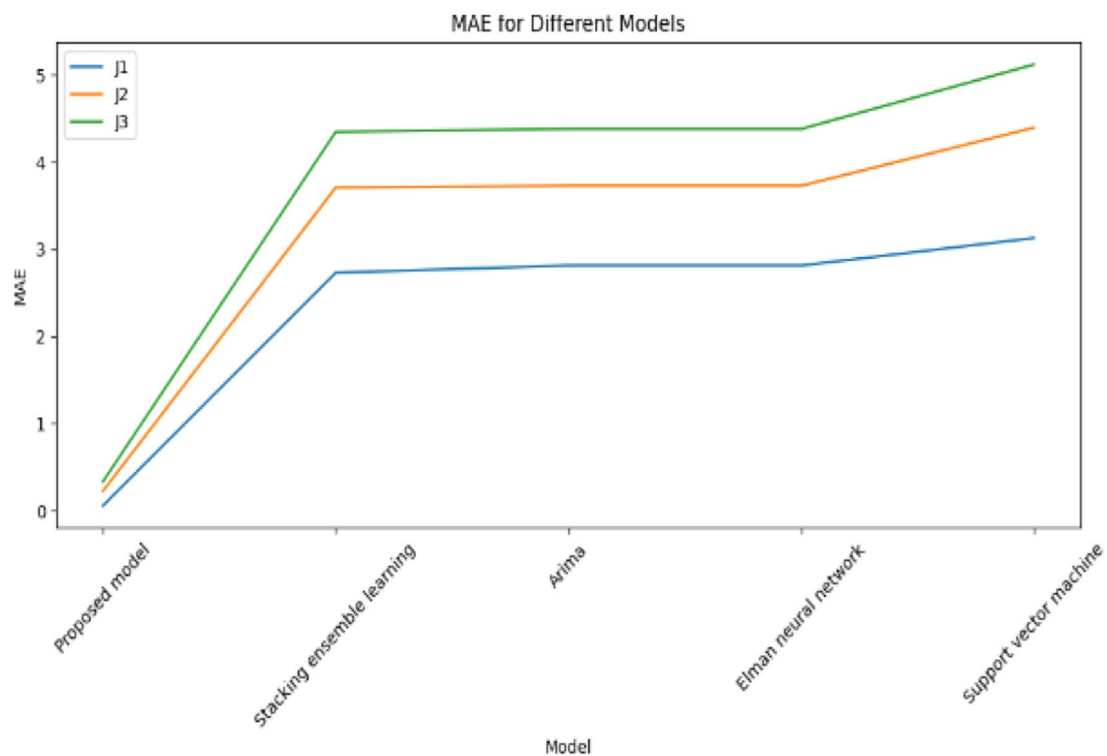


(b) MAE comparison with different models *Fedesoriano* dataset

Fig. 13 a RMSE comparison with different models using *Fedesoriano* dataset. b MAE comparison with different models *Fedesoriano* dataset

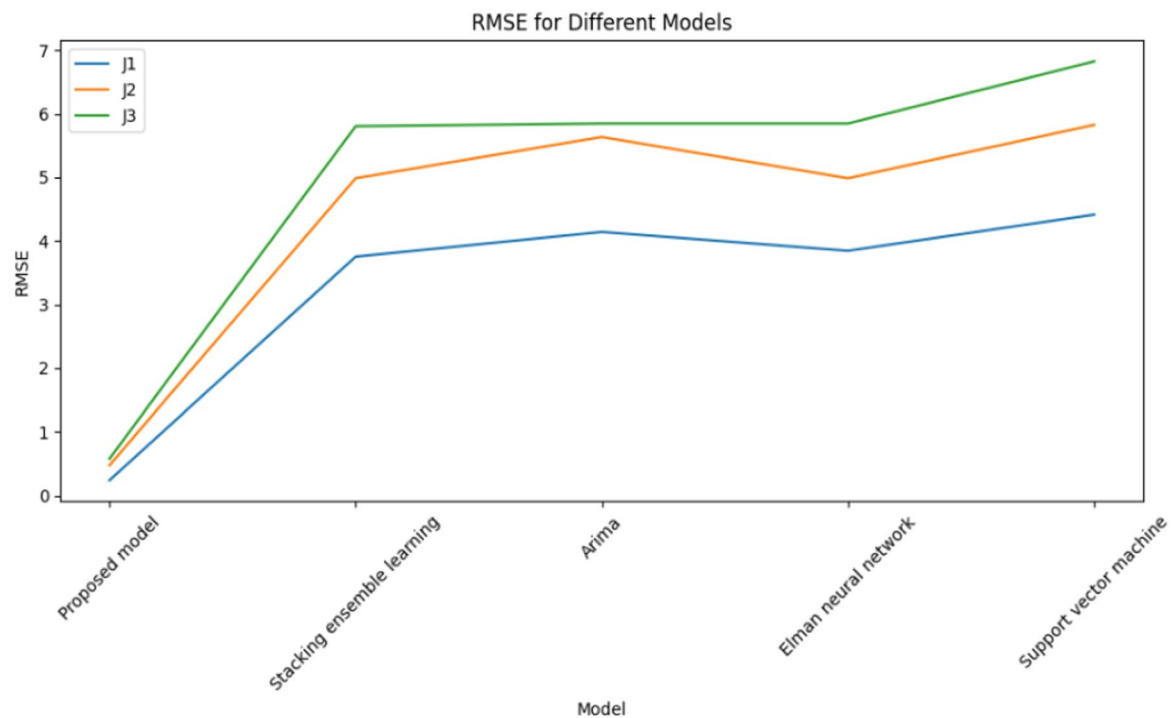


(a) MAPE Comparison with different models at J1, J2, J3 using *Fedesoriano* dataset



(b) MAE Comparison with different models at J1, J2, J3 using *Fedesoriano* dataset

Fig. 14 **a** MAPE comparison with different models at J1, J2, J3 using *Fedesoriano* dataset. **b** MAE comparison with different models at J1, J2, J3 using *Fedesoriano* dataset. **c** RMSE comparison with different models at J1, J2, J3 using *Fedesoriano* dataset



(c) RMSE Comparison with different models at J1, J2, J3 using *Fedesoriano* dataset

Fig. 14 (continued)

Table 8 Comparative analysis of RMSE and MAE for different model at 4 junction using *Fedesoriano* dataset

Model	RMSE				MAE			
	J1	J2	J3	J4	J1	J2	J3	J4
GRU	0.2458	0.5585	0.6061	1.0241	0.0604	0.3119	0.3673	1.0487
LSTM	1.0394	1.3541	1.4661	1.0189	1.0803	1.8335	2.1494	1.0381
CNN	0.2403	0.5389	0.6105	1.0086	0.0577	0.2904	0.3728	1.0179
GRU-LSTM	1.2863	1.1980	1.1843	1.5621	1.6541	1.4352	1.4007	2.4383
GRU-CNN	1.9863	1.7542	1.9042	1.9871	3.9462	3.0794	3.6256	3.9485
Extra Trees Regressor [51]	0.239889	0.502408	0.608794	1.046978	0.173067	0.391098	0.291559	0.741424
Light Gradient Boosting [51]	0.240309	0.482915	0.598730	1.010811	0.173558	0.376074	0.289985	0.709021
Decision Tree Regressor [51]	0.376209	0.708651	0.902702	1.321244	0.263177	0.537885	0.440593	0.931546
Hybrid CNN-GRU-LSTM	0.2391	0.4740	0.5775	1.0096	0.0572	0.2248	0.3334	1.0194

Table 9 Comparative analysis of RMSE, MAE and MAPE for different models at 3 junctions using *Fedesoriano* dataset

Model	RMSE			MAE			MAPE		
	J1	J2	J3	J1	J2	J3	J1	J2	J3
Proposed model	0.2391	0.4740	0.5775	0.0572	0.2248	0.3334	1.9012	1.8289	1.3470
Stacking ensemble learning [52]	3.752	4.986	5.805	2.730	3.708	4.347	15.136	20.898	25.638
Arima [52]	4.141	5.635	5.847	2.813	3.728	4.380	16.506	23.309	30.224
Elman neural network [52]	3.845	4.986	5.847	2.813	3.728	4.380	15.469	21.125	25.817
Support vector machine [52]	4.413	5.826	6.825	3.128	4.395	5.120	16.480	26.174	30.3336

Author contributions All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Vandana Singh, Dr. Sudip Kumar Sahana and Dr. Vandana Bhattacharjee. The first draft of the manuscript was written by Vandana Singh and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding No funding.

Data availability The datasets analysed during the current study are available in the ENGLAND dataset repository [<http://tris.highwaysengland.co.uk/detail/trafficflowdata#site-collapse-2011>], PeMS dataset repository [<http://pems.dot.ca.gov/>], P/CASTELLANA dataset repository [<https://github.com/MMH1997/CNN-BiLSTM-network>] and FEDESORIANO dataset repository [<https://www.kaggle.com/datasets/fedesoriano/traffic-prediction-dataset>].

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publications Not applicable.

Competing interests The authors declare no competing interests.

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References

1. Xing Z, Zhao S, Guo W, Guo X, Wang S, Li M, He H. Analyzing point cloud of coal mining process in much dust environment based on dynamic graph convolution neural network. *Environ Sci Pollut Res*. 2023;30(2):4044–61.
2. Xing Z, Zhao S, Guo W, Guo X. Geometric feature extraction of point cloud of chemical reactor based on dynamic graph convolution neural network. *ACS Omega*. 2021;6(33):21410–24.
3. Muhammad AN, Aseere AM, Chiroma H, Shah H, Gital AY, Hashem IAT. Deep learning application in smart cities: recent development, taxonomy, challenges and research prospects. *Neural Comput Appl*. 2021;33:2973–3009.
4. Vlahogianni EI, Karlaftis MG, Golias JC. Short-term traffic forecasting: where we are and where we're going. *Transp Res C*. 2014;43:3–19. <https://doi.org/10.1016/j.trc.2014.01.005>.
5. Li Y, Shahabi C. A brief overview of machine learning methods for short-term traffic forecasting and future directions. *SIGSPATIAL Special*. 2018;10(1):3–9. <https://doi.org/10.1145/3231541.3231544>.
6. Nagy AM, Simon V. Survey on traffic prediction in smart cities. *Pervasive Mob Comput*. 2018;50:148–63. <https://doi.org/10.1016/j.pmcj.2018.07.004>.
7. Zhou X, Zhang Y, Li Z, Wang X, Zhao J, Zhang Z. Large-scale cellular traffic prediction based on graph convolutional networks with transfer learning. *Neural Comput Appl*. 2022;34:5549–59.
8. Wang Z, Su X, Ding Z. Long-term traffic prediction based on LSTM encoder-decoder architecture. *IEEE Trans Intell Transp Syst*. 2021;22(10):6561–71. <https://doi.org/10.1109/tits.2020.2995546>.
9. Manibardo EL, Laña I, Lobo JL, Del Ser J. New perspectives on the use of online learning for congestion level prediction over traffic data. In: *International Joint Conference on Neural Networks (IJCNN)*, Glasgow, UK, 2020, p. 1–8. <https://doi.org/10.1109/IJCNN48605.2020.9207661>.
10. Williams B, Hoel L. Modeling and forecasting vehicular traffic flow as a seasonal arima process: theoretical basis and empirical results. *J Transp Eng*. 2003;129(6):664–72.
11. Zivot E, Wang J. Vector autoregressive models for multivariate time series. *Modeling financial time series with S-plus*. New York: Springer; 2006. p. 385–429.
12. Chen R, Liang C, Hong W, Gu D. Forecasting holiday daily tourist flow based on seasonal support vector regression with adaptive genetic algorithm. *Appl Soft Comput*. 2015;26:435–43.
13. Zafar N, Haq IU, Chughtai JUR, Shafiq O. Applying hybrid LSTM-GRU model based on heterogeneous data sources for traffic speed prediction in urban areas. *Sensors*. 2022;22(9):3348.
14. Modi Y, Teli R, Mehta A, Shah K, Shah M. A comprehensive review on intelligent traffic management using machine learning algorithms. *Innovative Infrastruct Solut*. 2022;7(1):128.
15. Zheng H, Li X, Li Y, Yan Z, Li T. GCN-GAN: integrating graph convolutional network and generative adversarial network for traffic flow prediction. *IEEE Access*. 2022;10:94051–62.
16. Tang J, Zeng J, Wang Y, Yuan H, Liu F, Huang H. Traffic flow prediction on urban road network based on license plate recognition data: combining attention-LSTM with genetic algorithm. *Transportmetrica A Transp Sci*. 2021;17(4):1217–43.
17. Kumar SV, Vanajakshi L. Short-term traffic flow prediction using seasonal ARIMA model with limited input data. *Eur Transp Res Rev*. 2015;7(3):1–9.

18. Shu W, Cai K, Xiong NN. A short-term traffic flow prediction model based on an improved gate recurrent unit neural network. *IEEE Trans Intell Transp Syst.* 2021;23(9):16654–65.
19. Chen C, Liu Z, Wan S, Luan J, Pei Q. Traffic flow prediction based on deep learning in Internet of vehicles. *IEEE Trans Intell Transp Syst.* 2020;22(6):3776–89.
20. Guo G, Yuan W. Short-term traffic speed forecasting based on graph attention temporal convolutional networks. *Neurocomputing.* 2020;410:387–93.
21. Li C, Xu P. Application on traffic flow prediction of machine learning in intelligent transportation. *Neural Comput Appl.* 2020;33(2):613–24. <https://doi.org/10.1007/s00521-020-05002-6>.
22. Wang H, Wei X, Yao J, Zhang Y. Traffic Flow Prediction Using Machine Learning Methods. In: 3rd International Conference on Machine Learning, Big Data and Business Intelligence (MLBDI), 2021. <https://doi.org/10.1109/mlbdi54094.2021.00014>.
23. Yin X, Wu G, Wei J, Shen Y, Qi H, Yin B. Deep learning on traffic prediction: methods, analysis, and future directions. *IEEE Trans Intell Transp Syst.* 2022;23(6):4927–43. <https://doi.org/10.1109/tits.2021.3054840>.
24. Johansson U, Boström H, Löfström T, Linusson H. Regression conformal prediction with random forests. *Mach Learn.* 2014;97:155–76.
25. Goodfellow I, Bengio Y, Courville A. Deep learning. Cambridge, MA: MIT Press; 2016.
26. Bruna J, Zaremba W, Szlam A, LeCun Y. Spectral networks and locally connected networks on graphs. In: Proceedings of International Conference on Learning Representations. 2014.
27. Rumelhart D, Hinton G, Williams R. Learning representations by back-propagating errors. *Nature.* 1986;323(6088):533–6.
28. Hochreiter S, Schmidhuber J. Long short-term memory. *Neural Comput.* 1997;9(8):1735–80.
29. Elman J. Distributed representations, simple recurrent networks, and grammatical structure. *Mach Learn.* 1991;7(2–3):195–225.
30. Meena G, Sharma D, Mahrishi M. Traffic prediction for intelligent transportation system using machine learning. In: 3rd International Conference on Emerging Technologies in Computer Engineering: Machine Learning and Internet of Things (ICETCE), Jaipur, India, 2020, p. 145–148. <https://doi.org/10.1109/ICETCE48199.2020.9091758>.
31. Wang H, Wei X, Yao J, Zhang Y. Traffic flow prediction using machine learning methods. In: 3rd International Conference on Machine Learning, Big Data and Business Intelligence (MLBDI), Taiyuan, China. 2021; p. 30–35. <https://doi.org/10.1109/MLBDI54094.2021.00014>.
32. Yu B, Li M, Zhang J, Zhu Z. 3D graph convolutional networks with temporal graphs: A spatial information free framework for traffic forecasting. *arXiv preprint arXiv:1903.00919.* 2019
33. Feng A, Tassioulas L. Adaptive graph spatial-temporal transformer network for traffic forecasting. In: Proceedings of the 31st ACM International Conference on Information & Knowledge Management. 2022. p. 3933–3937.
34. Zhang M, Li Y, Sun F, Guo D, Hui P. Adaptive spatio-temporal convolutional network for traffic prediction. In: 2021 IEEE International Conference on Data Mining (ICDM). IEEE; 2021. p. 1475–1480.
35. Zakarya M, Khan AA, Qazani MRC, Ali H, Al-Bahri M, Khan AUR, Ali A, Khan R. Sustainable computing across datacenters: a review of enabling models and techniques. *Comput Sci Rev.* 2024;52:100620.
36. Wei L, Yu Z, Jin Z, Xie L, Huang J, Cai D, et al. Dual graph for traffic forecasting. *IEEE Access.* 2019;20:19–20. <https://doi.org/10.1109/ACCESS.2019.2958380>.
37. Waikhom L, Patgiri R, Singh LD. Dynamic temporal position observant graph neural network for traffic forecasting. *Appl Intelligence.* 2023;53(20):23166–78.
38. Ali A, Zhu Y, Zakarya M. Exploiting dynamic spatio-temporal graph convolutional neural networks for citywide traffic flows prediction. *Neural Netw.* 2022;145:233–47.
39. Ali A, Ullah I, Shabaz M, Sharafian A, Khan MA, Bai X, Qiu L. A resource-aware multi-graph neural network for urban traffic flow prediction in multi-access edge computing systems. *IEEE Trans Consumer Electron.* 2024;70(4):7252–65.
40. Wen W, Xu D, Xia Y. A novel traffic optimization method using GRU based deep neural network for the IoV system. *PeerJ Comput Sci.* 2023;9: e1411. <https://doi.org/10.7717/peerj-cs.1411>.
41. Poonia P, Jain VK. Short-term traffic flow prediction: using LSTM. In: 2020 International Conference on Emerging Trends in Communication, Control and Computing (ICONC3), Lakshmangarh, India, 2020. p. 1–4. <https://doi.org/10.1109/ICONC345789.2020.9117329>.
42. Albawi S, Mohammed T, Al-Zawi S. Understanding of a convolutional neural network. In: Proceedings of the 2017 International Conference On Engineering And Technology (ICET), Antalya, Turkey, 21–23 August 2017. p. 1–6.
43. <http://tris.highwaysengland.co.uk/detail/trafficflowdata#site-collapse-2011>.
44. <http://pems.dot.ca.gov/>.
45. <https://github.com/MMH1997/CNN-BiLSTM-network>.
46. Fedesoriano. Traffic Prediction Dataset. Feb 2021. <https://www.kaggle.com/datasets/fedesoriano/traffic-prediction-dataset>
47. Rajalakshmi V, Ganesh Vaidyanathan S. Hybrid CNN-LSTM for traffic flow forecasting. In: Proceedings of 2nd International Conference on Artificial Intelligence: Advances and Applications: ICAIAA 2021 (p. 407–414). Singapore: Springer Nature Singapore; 2022.
48. Narmadha S, Vijayakumar V. Spatio-Temporal vehicle traffic flow prediction using multivariate CNN and LSTM model. *Mater Today.* 2023;81:826–33.
49. Fang Y, Qin Y, Luo H, Zhao F, Xu B, Zeng L, Wang C. When spatio-temporal meet wavelets: Disentangled traffic forecasting via efficient spectral graph attention networks. In 2023 IEEE 39th International Conference on Data Engineering (ICDE) (p. 517–529). IEEE; 2023.
50. Méndez M, Merayo MG, Núñez M. Long-term traffic flow forecasting using a hybrid CNN-BiLSTM model. *Eng Appl Artif Intell.* 2023;121: 106041.
51. Chaoura C, Lazar H, Jarir Z. Predictive system of traffic congestion based on machine learning. In: 2022 9th International Conference on Wireless Networks and Mobile Communications (WINCOM) (p. 1–6). IEEE; 2022.
52. Yin S, Liu H, Li Y, Tan J, Wang J. Multi-step traffic flow prediction using stacking ensemble learning model. In: Fifth International Conference on Traffic Engineering and Transportation System (ICTETS 2021) (vol. 12058, p. 638–643). SPIE; 2021.
53. Srivastava S, Sahana SK. A survey on traffic optimization problem using biologically inspired techniques. *Nat Comput.* 2019;16(67):1–15.